Shaurya Kishore Panwar

Indian — Artificial Intelligence + Robotics Engineer

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Education

University of Zurich — **Sept. 2024 – Present**
M.Sc. in Artificial Intelligence (Minors in Data Science) — *Zurich, Switzerland*

Delhi Technological University — **Dec. 2020 – Jul. 2024**
B.Tech. in Engineering Physics (Minors in Computer Science), ***CGPA: 8.32/10*** — *Delhi, India*

St. Xavier's Senior Sec. School — **Jan. 2018 – Mar. 2020**
Central Board of Secondary Education, Percentage: ***89.4%*** — *Rohini, Delhi, India*

Languages:

- English - C1
- German - A2/B1
- Hindi - Native
- French - A1

Relevant Coursework

- Comp. Vision for Robotics
- Algorithms Design & Analysis
- Deep Learning
- Vision-Based Drone Flight
- Reinforcement Learning
- Instrumentation & Control
- Mixed Reality
- Large Language Models
- Software Engineering

Technical Skills

Languages: Python, C++, Java, Arduino, Bash Scripting, LaTeX

Machine Learning: Deep reinforcement learning, dzeep learning including CNNs, RNNs, VAEs, GANs, Transformers, NeRFs; Machine Learning including SVM, KNN, Decision Trees, Bayes

Framework and Libraries: Linux, PyTorch, TensorFlow, OpenCV, ROS (Robot Operating System), SciKit-learn, AutoWare, Gazebo, Pandas, Open3D, PyCeres, NumPy, Matplotlib, SQL

Tools: Git, Docker, postgres SQL, Anaconda

Publications

- **Shaurya K. Panwar**, Anil S. Parihar *"Elevating Semantics: Fine-Grained Semantic Grounding for Enhanced Aerial Vision and Language Navigation"* Bachelor's Thesis at DTU

Experience

OpenCV - Google Summer of Code (GSOC25) Contributor — **May. 2025 – Sept. 2025**
Advisor: Gary Bradski (OpenCV) , Gholamreza Amayeh (Tesla) — *Remote*

- Repository- github.com/KlrShaK/opencv-SimpleSLAM
- Development of real-time Visual SLAM from scrath in python, with loop closure and global pose-graph optimization(PyCeres) with SOTA learned features (ALIKED + LightGlue).

Semester Project - ETH Zurich & Uni. Zurich — **May. 2025 – Present**
Advisors: Prof. Dr. Marc Pollefeys, Dr. Dániel Béla Baráth , Prof. Dr. Manuel Günther — *Zürich, Switzerland*

- Interdisciplinary research on Language-driven localization and place recognition in 3D environments. Built Large Vision-Language Foundational Models(LVLMs) based retrieval & grounding pipelines for embodied agents.

Visiting Researcher - Bern University of Applied Science — **Jun. 2023 – Aug. 2023**
Advisor: Prof. Marcus Hudrich — *Bern, Switzerland*

Worked on *"No GPS Drone"* project to develop system for visual navigation, removing reliance on GPS.

- Enhanced segmentation accuracy from F1 score of 64.11 to 88.38. Redesigned ML network using transformer based decoder(UNetFormer). Expanded training dataset with augmentation and leveraged transfer learning techniques to optimize performance. While maintaining fast inference rate(67 FPS) on Jetson Orin Nano.
- Led the selection and integration of critical sensors, including cameras, altimeters, IMU, etc. Developed a synchronized data acquisition pipeline with data filtering and augmentation for seamless integration into the drone.

Student Researcher - Delhi Technological University **Jan. 2022 – Aug. 2024**
Advisor: Prof. Anil Singh Parihar *Delhi, India*

- Interdisciplinary research in Vision-Language Navigation at intersection of Computer Vision, Deep Reinforcement Learning and Natural Language Processing, specifically with applications in embodied navigation.
- Authored a significant work in the field of Vision-Language Navigation(VLN), **Ranking #5 in VLN Leaderboard.** (results under Team Name *"MLR-Lab-DTU"*)

Projects

Deep RL Vision-based Drone Control | *Python, C++, TensorFlow, ROS2, OpenCV* **Sept. 2025**

- Complete Report Link
- Shipped a RL controller (PPO) for autonomous drone target-following (under partial observability); Successful real-world deployment with tight altitude tracking (-0.10 m mean error) and safe stand-off control.
- Engineered perception-to-control stack (double-sphere bbox sim, FoV rewards, command smoothing) with domain randomization; improved sim-to-real robustness.

Autonomous Systems Lead - Team Defianz-DTUSDC **Nov. 2020 – Jul. 2023**
Advisors: Prof. Anil S. Parihar & Prof. Qasim Murtaza *Delhi, India*

Developing Autonomous Framework for Open-Wheeled Formula Student Driverless Race car, one of the first driverless team in India. Achieved 8th Position worldwide in FS Italy 2022

Perception Subsystem | *Python, TensorFlow, ROS, Darknet-YOLOv4*

- Developed high-speed CV pipeline for traffic cone detection, localization, and classification. Enhanced accuracy by 20% using image augmentation, transfer learning, and custom synthetic data generation from limited initial dataset.
- Developed system for high speed 3D object detection in pointclouds from LIDAR data using PointPillar technique.

Leadership / Extracurricular

Team Defianz-DTUSDC Racing **Jan. 2022 – Jul. 2024**
Team Manager & Autonomous System Lead *Delhi Tech. University*

- Personally responsible for acquiring Title Sponsorships worth over $5000 from a tech giant like NXP Semiconductors.
- Supervised and managed over 40+ people in the team.
- Served as Autonomous Systems Officers (ASO), certifying the vehicle and its HV(High Voltage) + AS systems for safety, demonstrating expertise in industrial systems and compliance standards

Achievements and Awards

- **Google Summer of Code 2025** Contributor for **OpenCV** – Selected from over 400 people to implement and add a SLAM framework into current OpenCV stack.
- **ThinkSwiss Scholarship Awardee** at Bern University of Applied Sciences, Switzerland.
- **8th** in **FS Italy 2022** Engineering Design Event
- **3rd** in **NXP AIM India Smart Car Race Challenge** and Best Car Model Award from over 400 teams nationally
- **2nd** in **Formula Bharat FSEV** Software and Integration Report.